# BUSINESS REQUIREMENTS DOCUMENT

## Travel Planning & Booking Application

*Version 1.0 | 9 October 2026*

Document purpose: Define the business problem, objectives, users, scope, and expected capabilities of a travel application before detailed design and development begin.

## 1. Project Overview

The Travel Planning & Booking Application will help travellers discover destinations, plan trips, organise itineraries, estimate expenses, and access accommodation or transport booking options in one place. The first release is intended as a practical MVP (minimum viable product); actual reservations and payments depend on integration with suitable third-party providers.

## 2. Business Problem

Travellers often use multiple websites and apps to research destinations, compare options, organise daily plans, and track trip expenses. Information can become scattered, making trip preparation time-consuming and difficult to manage.

## 3. Business Objectives

- Provide one convenient place to discover destinations and build trip plans.
- Allow users to create, edit, and view day-by-day itineraries.
- Help users estimate and track a trip budget.
- Present useful destination information, including attractions and travel tips.
- Provide clear links or integrations for accommodation and transport options.
- Offer an accessible, mobile-friendly experience that reduces planning effort.

## 4. Target Users / Stakeholders

- Guest visitor: explores destinations and public travel information.
- Registered traveller: saves destinations, creates trips, manages itineraries, and tracks budgets.
- Administrator: manages destination content, categories, reports, and inappropriate or outdated information.
- External providers: map, weather, transport, accommodation, or payment services integrated where required.
- Project team: product owner, designer, developer, and tester.

## 5. Functional Requirements

| ID / Feature | Requirement |
|---|---|
| **FR-01: Registration and login** | Users can register, log in, log out, and securely manage their profile. Guests may browse public content. |
| **FR-02: Destination discovery** | Users can browse and search destinations by name, |

| | |
|---|---|
| | region, category, or interests. |
| **FR-03: Destination details** | The app displays available descriptions, images, attractions, suggested activities, travel tips, and estimated costs. |
| **FR-04: Trip creation** | Registered users can create a trip with a title, destination, start date, end date, and number of travellers. |
| **FR-05: Itinerary management** | Users can add, edit, reorder, and remove activities or places by day. |
| **FR-06: Budget management** | Users can set a budget and record estimated or actual expenses by category, such as transport, stay, food, and activities. |
| **FR-07: Saved places** | Users can bookmark destinations and attractions for later. |
| **FR-08: Maps and directions** | Where a map provider is available, users can view locations and open directions. |
| **FR-09: Booking information** | The app can show or link to third-party transport and accommodation options. In-app confirmation is only included if provider APIs support it. |
| **FR-10: Notifications** | The app may send trip reminders or itinerary updates when notification services are configured. |
| **FR-11: Admin content management** | Administrators can add, edit, publish, or remove destination and attraction information. |
| **FR-12: Feedback and support** | Users can submit feedback or report incorrect destination information. |

## 6. Non-Functional Requirements

- Usability: Common tasks should be understandable to first-time users, with clear navigation and helpful validation.
- Responsive design: Core screens should work on mobile phones, tablets, and desktop browsers.
- Performance: Main screens and searches should load promptly under normal network conditions.
- Security: Passwords must not be stored in plain text; access to private trips and profile data must be controlled.
- Privacy: Collect only necessary personal data and explain how it is used.
- Reliability: Show clear errors when a third-party API is unavailable; do not present failed bookings as confirmed.
- Maintainability: Keep destination data, trip logic, and external integrations organised so they can be updated independently.
- Accessibility: Use readable contrast, labels, keyboard support, and meaningful alternative text where applicable.

## 7. Business Rules

- A user must be authenticated to save a personal trip or modify their itinerary.
- A trip end date must be the same as or later than its start date.
- A budget and expense amount cannot be negative.
- Only authorised administrators may publish or remove shared destination content.
- A third-party booking is confirmed only when the provider returns a valid confirmation; a link-out is not a booking.
- Prices, opening hours, weather, visa guidance, and availability may change and should display source or last-updated information where possible.

# 8. Project Scope

### In scope for the proposed MVP

- Account creation and login.
- Destination listing, search, and detail pages.
- Personal trip creation and day-by-day itinerary editing.
- Basic budget and expense tracking.
- Saved destinations and attractions.
- Basic admin management of destination content.
- Responsive user interface and basic error handling.

### Potential later-phase features

- Live flight, train, bus, and hotel search through provider APIs.
- In-app payments, refunds, and reservation management.
- Collaborative trip planning with friends.
- AI-generated itineraries and personalised recommendations.
- Offline maps or downloadable itineraries.
- Multi-language and multi-currency support.

### Out of scope for the initial MVP

- Operating transport or accommodation services directly.
- Guaranteeing third-party prices, schedules, availability, or entry requirements.
- Providing emergency response or acting as an insurer.
- Processing bookings or payments unless a provider integration is explicitly approved.

# 9. Assumptions

- Users have access to a smartphone or web browser and an internet connection for online features.
- Destination content can be entered by administrators or sourced from permitted APIs.
- Third-party map, weather, and booking APIs may require keys, usage limits, or paid plans.
- The initial release can use booking links if direct booking APIs are unavailable.
- The project team will confirm the launch geography and supported currencies before implementation.

# 10. Constraints and Dependencies

- Budget, team size, and delivery deadline have not yet been confirmed.
- Features relying on external providers depend on their API access, pricing, terms, and uptime.
- Personal data and payments must be handled according to applicable laws and provider requirements.
- Accurate live information depends on the freshness and reliability of third-party data.

# 11. High-Level User Flow

Open app → Browse/search destinations → View destination details → Register/log in → Create trip → Add dates and travellers → Build itinerary → Set budget → Save trip → Optionally open external booking links → Review trip plan

## 12. Success Criteria

- A user can create an account and sign in successfully.
- A user can find a destination using search or browsing.
- A registered user can create, edit, and reopen a saved trip.
- A user can add itinerary items and see them grouped by day.
- A user can record expenses and view the remaining budget.
- An administrator can update destination content.
- Errors from unavailable external services are communicated clearly and do not create false booking confirmations.

## 13. Risks and Mitigations

| Risk | Mitigation |
|---|---|
| **Outdated destination information** | Show update dates and provide an admin workflow to review content. |
| **Third-party API limits or cost** | Confirm provider terms early; use a modular integration and fallback links. |
| **Scope becoming too large** | Deliver the MVP first and treat live bookings/AI features as later phases. |
| **Privacy or account security issues** | Use established authentication practices, access controls, and minimal data collection. |
| **Unclear booking status** | Clearly distinguish information links, pending requests, and provider-confirmed bookings. |

## 14.Resources and Technologies Required

- Frontend: HTML, CSS, JavaScript, and React for the user interface.
- Styling: Tailwind CSS or regular CSS for responsive page design.
- Database: MySQL
- Maps: OpenStreetMap, or Google Maps API
- Design: Figma for wireframes and UI planning
- Version control: Git and GitHub.
- Testing: Browser developer tools and manual functional testing.

## 15. Acceptance and Approval

This BRD should be reviewed by all the members of the project before detailed software requirements, UI/UX design, database design, and implementation begin.